Dept. of Software Engineering

A Markerless AI Goniometer for Ankle Dorsiflexion Measurement

Ron Kroitoro and Sheli Zisman
Supervisors: Dr. Naomi Unkelos-Shpigel and Dr. Einat Ravid
Department of Software Engineering, Braude College of Engineering, Karmiel, Israel

MOTIVATION

Ankle dorsiflexion: bringing the toes toward the shin -enables forward propulsion in gait. Restriction is linked to injury, aging, fall risk, and impaired quality of life.

The universal goniometer has poor inter-rater reliability - most clinicians abandon it for visual estimation.

≤5°
clinical accuracy requirement

1.7°
max. implied angle error (geometric estimate)

CLINICIAN NEEDS ASSESSMENT

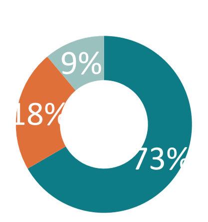
Interviews with 2 orthopedic specialists + survey of 11 practitioners (8 physiotherapists, 2 surgeons, 1 podiatric surgeon; all 5+ years experience).

> *"We're so busy that we often just estimate the angle 'by eye.' Even when we do use the goniometer, if you move it just a tiny bit, the result changes completely."*
>
> **— Senior Orthopedic Surgeon, Head of Department**

> *"It is very difficult to accurately identify the correct measurement point, especially when the foot is swollen... this often results in inconsistent measurements and a form of angle 'guessing.'"*
>
> **— Orthopedic Resident**

Six recurring themes: clinical consequences · rehabilitation impact · perceived inaccuracy · goniometer abandonment · desired simplicity · data integration needs.

Survey Respondents (n = 11)



5+ years
clinical experience (all respondents)

7 open-ended questions
on consequences, usage, features & adoption intent

CLINICAL INTERVIEW INSIGHTS

1. Practical difficulties: landmark placement, device positioning, examiner variability
2. Workflow needs: fast capture, simple operation, minimal disruption
3. System requirements: clear ROM output, confidence score, session comparison
4. System design: two-image input, auto landmark detection, angle calculation, measurement report

LIMITATIONS & FUTURE WORK

Limited to healthy young adults. Future: larger dataset, improved robustness and clinical validation

SYSTEM OVERVIEW

Input: 2 smartphone photos (neutral + max. dorsiflexion). No markers or specialist equipment required.

3 keypoints detected automatically (tibial midline, lateral malleolus, 5th metatarsal). ROM is calculated as the absolute angular difference between the resting and maximum-dorsiflexion positions.

SIX-STAGE PROCESSING PIPELINE

1. **Dual-image capture** — Neutral position + maximum dorsiflexion
2. **Image upload** — Web interface → REST API → backend server
3. **DeepLabCut inference** — ResNet-50 model returns 3 keypoints + confidence
4. **Confidence filtering** — Sub-cutoff predictions flagged as uncertain
5. **Angle computation** — Vector-based geometry → ROM delta
6. **Result display**— Keypoint overlay + calculated ROM value

MODEL: DEEPLABCUT + RESNET-50

Custom DeepLabCut model trained to detect 3 ankle landmarks: tibial midline, lateral malleolus, 5th metatarsal.
A ResNet-50 backbone extracts robust visual features for anatomical landmark localization. Pretrained on ImageNet.
Detected coordinates fed into vector-based angle computation → final ROM delta.

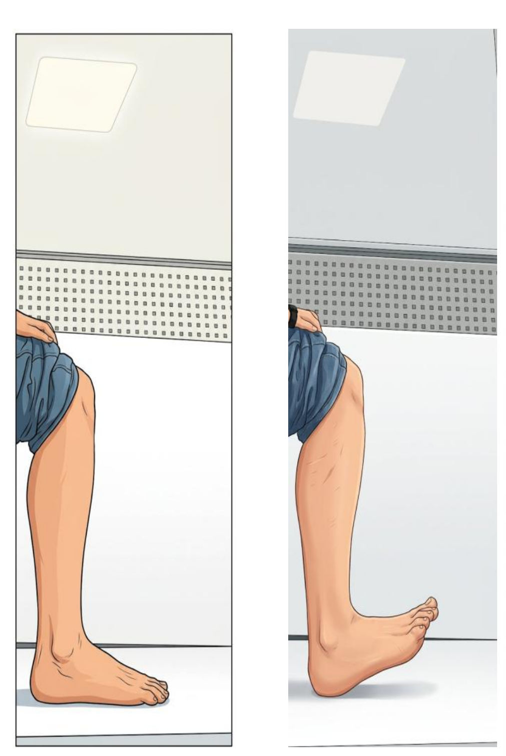
USER-CENTERED CLINICAL DESIGN

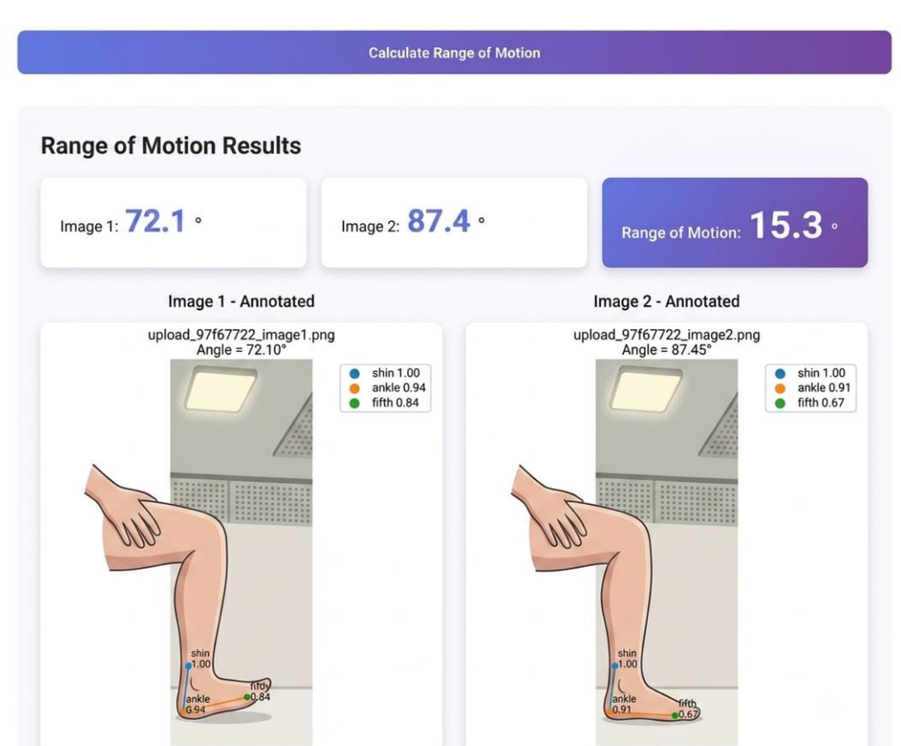
Simple image capture
2 smartphone photos. No markers or specialist equipment.

Visual verification
Detected landmarks are displayed on the image, allowing the user to verify the model output.

Clear ROM result
The calculated range-of-motion value is presented in a clear numerical format.

Confidence indication
Low-confidence landmark predictions are flagged as uncertain.





DATASET & TRAINING

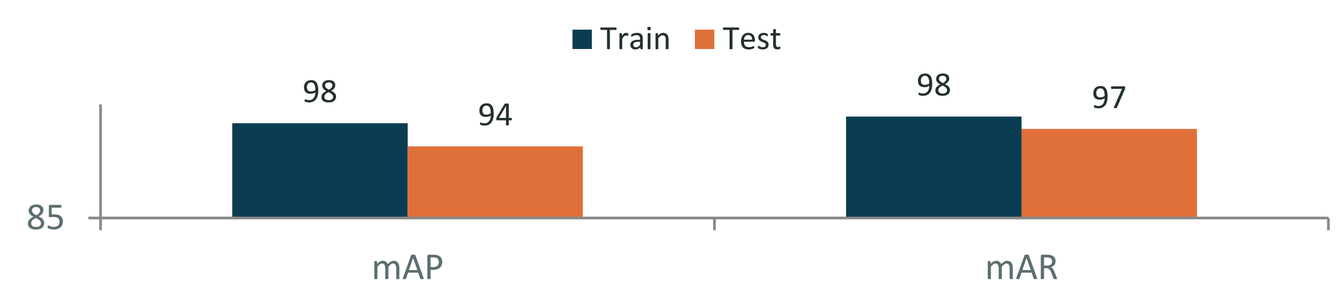
16 healthy adults (10M, 6F; mean age 25) — 369 images. iPhone 14, fixed tripod, 1.2 m distance, standardized lighting.

Parameter	Value
Keypoints per image	3 (tibial midline, lat. malleolus, 5th metatarsal)
Annotation method	Manual, single trained annotator
Dataset split	80% training / 20% testing (295 / 74 images)
Data separation	No overlap between splits

MODEL EVALUATION RESULTS

Metric	Train	Test
RMSE, px (all preds.)	6.07	8.05
RMSE, px (p-cutoff)	5.27	7.52
mAP (%)	97.54	94.48
mAR (%)	98.41	96.76

Keypoint Detection Performance: Train vs. Test (%)



94.48%
test mAP — detection performance

7.52 px
test RMSE at p-cutoff

PLANNED CLINICAL VALIDATION

Dual-reference design:
Universal goniometer + custom mechanical protractor device.

Participants:

Healthy adults aged 22–60; multiple repetitions per session, all 3 methods applied simultaneously.

Primary metric:

MAE vs. goniometric reference. Secondary: detection success rate, processing time, user completion time.

CONCLUSIONS

- Practical 2-photo approach for measuring ankle dorsiflexion ROM - no specialist equipment needed.
- Automatic detection reduces dependence on manual goniometer placement and examiner variability.
- Confidence scores flag uncertain detections - clinician can review or repeat.
- Designed for clinical and home-based follow-up; supports longitudinal ROM tracking.